Group 2: Written justification

Building Age Classification · ZAS Data Science Regional Hackathon 2026 · 292 words (limit 300)

Model design. Building age leaves three traces in Landsat data: material (spectral shape), surroundings, and, for buildings erected after 1984, a construction event. Our features target each: indices and robust statistics for material, neighbourhood averages for surroundings, and one- and two-break change-points for construction; break strength separates pre- from post-1984 classes in both cities. We compared six algorithms with spatial-block cross-validation, because random folds leak near-identical neighbours (0.736 random vs 0.652 spatial for the same model). Gradient-boosted trees performed best; their average with Random Forest is our Madrid model.

Transfer strategy. Our ablation showed features alone could not lift Amsterdam above ≈ 0.60 at n = 25: the obstacles are domain shift (Amsterdam is darker and greener) and a different class mix. We therefore standardise each city separately, transfer only city-independent features, and adapt with methods that reuse Madrid's knowledge: logistic recalibration of CatBoost probabilities, LightGBM retrained on Madrid plus heavily weighted Amsterdam samples, and self-training with confident pseudo-labels from unlabelled Amsterdam pixels. We average them, correct the class prior with EM only when n ≥ 25, and smooth probabilities spatially (tuned on Madrid only). Every upgrade was tested on identical draws; CORAL, calibration and TabPFN were rejected.

Interpreting the scores. With 5 labels per class (0.654 ± 0.021) Amsterdam already matches Madrid's cross-validated score, because self-training exploits thousands of unlabelled pixels. At n = 25 (0.687) the class-mix correction becomes reliable and adds about 0.02. Gains continue to n = 200 (0.757) with very small error bars, while prototype and recalibration plateau near 0.63–0.64. A final pre-registered check (labels from one half of Amsterdam in 2 km blocks, scores only on the other half) gives 0.710 versus 0.712, with no drop beyond 1 km from labels: the gain is not neighbour-copying.